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APEX PRE-TRADE COMPLIANCE GATE | SEC RULE 206(4)-7 PROTECTION

LIVE GATE | 10 / 10 VERIFIED

+$375,500 Saved | $0 Out-of-Pocket Loss Paid by the Firm

Checking parent-company links, split ratings, energy sub-limits, and deal footnotes before a trade executes.

+$375,500	$0 Paid	-$277,500	$10.70M
Unwind Losses & Costs Saved	Firm Loss With Our Gate	Loss & Costs Without Gate	10 / 10 Orders Screened

1. The Area I Explored & Three Things That Surprised Me

Domain Discovery

When I interned as a Software Engineer at **Apollo Global Management**, I built data delivery pipelines and dashboards across **75,000+ holdings and 20,000+ trade records**, along with a natural-language query assistant for portfolio data. That experience got me interested in a deceptively simple question: *what actually happens in the 60 seconds between a portfolio manager clicking 'Buy' and the order hitting the market?* When I dug into **SEC Rule 206(4)-7** and real fund rulebooks (Investment Policy Statements and Form N-1A filings) for this project, three things genuinely surprised me:

1. **A rule on Page 12 is defined on Page 2:** In a fund rulebook, a limit looks straightforward at first glance -- for example, *'No more than 5.00% of the fund in any single company'* (Section 3.01a). But if you flip back to Section 1.04, 'Single Company' is redefined to include every majority-owned subsidiary sharing a parent company. And if S&P rates a bond BBB- (Investment Grade) while Moody's rates it Ba1 (High-Yield), Section 2.02b says the higher rating wins. If you read the limit paragraph without pulling the definition paragraph, you get the math wrong.
2. **Holding a stock above 5% is legal, but buying $1 more is illegal:** If NVIDIA rallies and grows to 5.15% of the fund on its own (passive market drift), Section 3.03 lets the fund keep holding it without forced selling. However, executing even a $50,000 active BUY order while NVIDIA sits above 5.00% is an immediate regulatory breach.
3. **Why neither SQL rules nor raw LLMs can do this alone:** Traditional database checks are great at summing columns, but they cannot read a 3-sentence deal footnote warning that a rail company gets 22% of its revenue from mixed coal corridors, or that a Cayman Islands trust is secretly backed by an oil-and-gas swap basket. On the flip side, pasting 75,000 holdings rows into an LLM burns ~3.6 million tokens, leaks private custodian account numbers, and routinely fumbles parent-company joins and basis-point math.

2. The Problem We Solve & Why Bad Trades Cost Real Money

SEC Rule 206(4)-7

> **Why It Matters in Plain Dollars:** If a non-compliant trade slips through and gets executed, the fund breaks **SEC Rule 206(4)-7**. The fund is forced to undo (unwind) the trade in the open market. If the bond or swap is sold at a loss, **the investment firm must pay for that loss out of its own pocket** -- typically $30,000 to $500,000+ per trade error in bid-ask slippage, swap termination penalties, and legal audit fees.

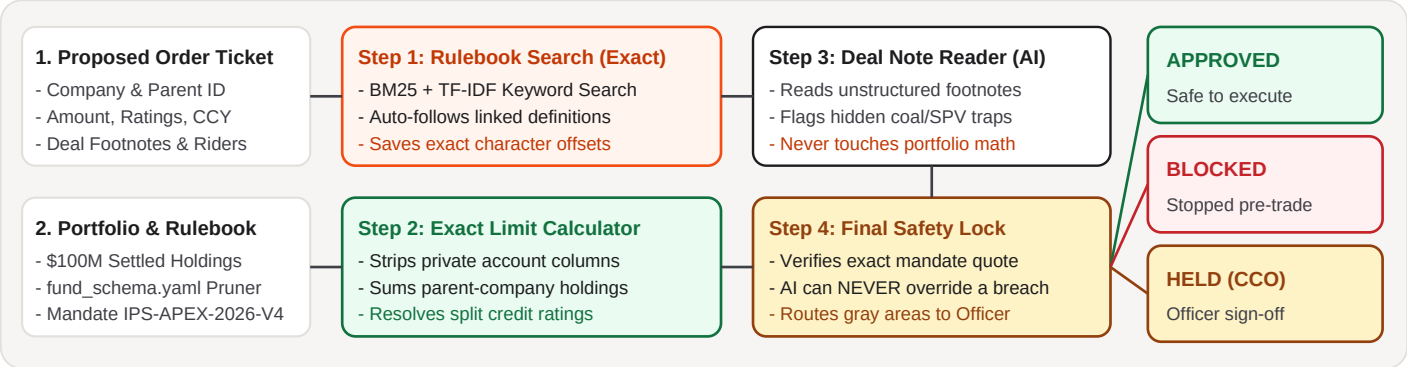
- **Who uses this & how often:** Pre-trade compliance officers and middle-office desks at multi-asset credit and equity funds ($500M to $50B AUM), screening **50 to 300+ proposed allocation tickets every trading day**.
- **What it costs today:** Checking a complex bond or swap across PDF rulebooks and side letters takes **15 to 30 minutes of manual lookup**. Rushing causes missed parent-group breaches (forcing costly market unwinds); being overly cautious causes false alarms on split-rated bonds like Ford Motor Credit (costing $80k+ in missed execution spreads).
- **Why now:** As private credit and custom institutional accounts grow, firms manage hundreds of bespoke client rulebooks. Hard-coding every new prose side letter into legacy SQL engines takes 3 to 6 weeks. Combining exact Python portfolio math with a constrained AI footnote reader cuts verification to seconds with zero math hallucinations.

What Is In Scope (Pre-Trade Gate)	What Is Explicitly Out of Scope
- Exact before-and-after trade % math across direct companies and parent groups (Sec 1.04 / 3.01). - Split credit rating tie-breakers (Sec 2.02b), High-Yield Energy sub-limits (3.02b), and foreign government bond rules (4.01b). - Reading deal footnotes for hidden coal freight (5.01b) and offshore SPV swaps (5.02), plus tracking **Losses Paid vs. Payments Saved**.	- Broker order-book routing, execution slicing, and FIX protocol market connectivity. - Post-trade T+1 custodian bank wire settlement and cash reconciliation. - Predicting stock/bond price returns or rewriting legal contracts automatically.

2.1 How the System Works: Exact Math First, AI Reading Second

4-Step Pipeline

The core design principle of our system is simple: **never let a language model do portfolio arithmetic, and never expect a SQL query to read a legal footnote**. In src/agent_graph.py, every proposed trade passes through four distinct steps where exact calculation and AI reading are kept strictly separate:



2.2 Where Knowledge Lives & How Rules Stay Up to Date

System Boundaries

System Step	Exact or AI?	Where Knowledge Lives	Why It Works & How We Verify It
1. Rulebook Search (src/retriever.py)	**100% Exact (BM25 + Graph)**	Versioned markdown rulebook (fund_mandate_ips.md)	Finds exact section numbers and automatically pulls linked definitions (e.g., Sec 3.01 -> Sec 1.04) with character offsets.
2. Limit Calculator (deterministic_engine.py)	**100% Exact (Python Math)**	holdings_nport.json + fund_schema.yaml	Strips private account numbers locally (99.993% token cut) and calculates exact parent-group, credit, and currency percentages.
3. Footnote Reader (src/agent_graph.py)	**Probabilistic (OpenRouter LLM)**	Retrieved rulebook clauses + deal supplement notes	Reads unstructured deal prose to spot gray areas (mixed coal rail corridors in Sec 5.01b or opaque Cayman SPVs in Sec 5.02).
4. Final Safety Gate (src/agent_graph.py)	**100% Exact (State Lock)**	verify_verbatim_quote() + Asymmetric Lock	Checks that the cited quote exists character-for-character in the rulebook. The AI can never turn a math breach into an approval.

- **What happens when a rule changes or expires:** No rules or limits are baked into model weights. Every threshold lives in fund_schema.yaml and fund_mandate_ips.md (tagged with Document ID IPS-APEX-2026-V4). When the board updates a cap or adds a client side letter, updating the file immediately updates Step 1 and Step 2 with zero retraining. Past trade logs keep the exact Mandate ID that was active when the trade ran.
- **Adapting to another country (e.g., Europe UCITS or India SEBI):** Our prototype enforces US SEC 1940 Act rules. Moving to European UCITS funds (which enforce the '5/10/40' rule: positions over 5% cannot sum to more than 40% of the fund) only requires swapping the rulebook file and adding a 4-line summation check in Step 2.

2.3 The 4-Dashboard Prototype (Plain-English Orange Edition)

Frontend Design

Compliance officers should not have to read raw JSON or backend code on screen. Modeled after the **AEGIS** visual system and **Emil Kowalski's** motion engineering principles, our Streamlit app (app.py) uses a **Warm Electric Orange (#F04E14)** theme and organizes the workflow into four clean, plain-English dashboards:

01 - Pre-Trade Gate	Features the signature 'Needs You' Split-Card with 30px tabular numerals (*Before trade -> After trade*), a 2-segment overshoot bar showing the exact red breach segment past the limit tick mark, a 4-step pulsing checkpoint stepper, a conic-gradient score ring, and a What-If simulator.
02 - Losses vs. Savings	Compares Firm P&L **Without Pre-Trade Gate (-$277,500 loss/cost)** vs. **With Our Gate ($0 loss paid, +$375,500 saved)**, breaking savings down across forced unwind slippage avoided (+$167,000), false alarms prevented (+$166,500), and SEC review fees saved (+$42,000).
03 - All Activity & Protection	A dedicated visual Bento dashboard across all 10 orders ($10.70M screened: $3.90M safely executed vs. $6.80M stopped/held), complete with an outcome donut ring, interactive decision filters, inline SVG icons, and per-trade P&L protection strips.
04 - Fund Portfolio ($100M)	Displays the $100.0M settled Form N-PORT portfolio ($82.35M across 12 holdings + $17.65M cash) with parent-group concentration bars, 5.00% limit markers, and passive market-drift indicators (e.g., NVIDIA at 5.15%).

3.1 Four Approaches We Tested and Rejected

Architectural Trade-Offs

Before settling on our 4-step hybrid design, we evaluated four common ways teams try to build compliance tools and rejected each one for specific, practical reasons:

Rejected Approach	Why It Fails on Real Institutional Trades
1. One Big LLM Call Over All Holdings + Rulebook	Sending 75,000 holdings rows (~3.6M tokens) to an LLM costs $10-$30 per trade check, takes 30+ seconds, exposes private custodian bank codes to external APIs, and routinely miscalculates parent-company sums and 5.00% vs. 5.01% boundary math.
2. Pure Vector Embedding Search (Dense RAG Only)	Legal rulebooks rely on exact section numbers (Section 3.01(b)) and rating symbols (Ba1, BBB-). Dense embeddings blur negations and miss multi-hop definitions (pulling Section 3.01 without pulling Section 1.04). Our BM25 + TF-IDF + Cross-Reference Graph fixes both.
3. 100% Hard-Coded SQL Rules (No AI Reader)	Fails silently on **TC-09 and TC-10**: when database columns look clean (coal_revenue = 4.2% <= 10%, rating = A-), SQL auto-approves the trade and misses the footnote revealing 22% mixed coal rail freight or an undisclosed Cayman swap basket.
4. Letting an LLM Write SQL Queries on the Fly	If a Text-to-SQL agent forgets GROUP BY ultimate_parent_id on just one run out of a hundred, a $1.3M subsidiary breach (like Florida Power & Light under NextEra Energy) slips through. Compliance math must be locked in tested code, not generated dynamically.

3.2 The Weakest Point in Our Design (And What Would Break It)

Honest Fragility Analysis

> **Our Biggest Vulnerability -- Stale Parent-Company Links in the Security Master (ultimate_parent_id):**
> Step 2 catches parent-subsidiary breaches (like knowing NextEra Energy owns Florida Power & Light) by grouping holdings on ultimate_parent_id. That math is 100% exact, but it trusts that the upstream corporate tree table is up to date.

- **How to break it (Scenario 1 -- Same-Day Spin-Off or New Financing Shell):** Suppose an existing portfolio company creates a brand-new Delaware financing subsidiary this morning to issue bonds, and our data vendor has not linked the new subsidiary's ultimate_parent_id yet. Step 2 will see 0% existing exposure and approve the trade unless the deal footnote explicitly names the parent company for Step 3 to catch.
- **How to break it (Scenario 2 -- Chained Side-Letter Waivers):** Our cross-reference expander follows 1-hop links (e.g., Section 3.01 -> Section 1.04). If an institutional client signs three sequential side-letter riders that modify each other conditionally (*'Rider 3 pauses Rider 1 during Q4 unless High-Yield exceeds 12%'*), a 1-hop lookup can miss the interaction between Rider 3 and Rider 1.

3.3 Managing Real-World Risks & Human-in-the-Loop Governance

Risk & Liability

Risk Category	What Happens If Something Goes Wrong	How Our Architecture Protects the Firm
1. Unequal Cost of Errors (False Positive vs. False Negative)	Pausing a good trade costs 2 minutes of officer review. Approving a bad trade triggers SEC Rule 206(4)-7 forced unwinds and $50k-$500k+ out-of-pocket firm loss.	**One-Way Safety Lock:** Step 4 is biased strictly toward safety. The AI can escalate a clean math check to HELD FOR OFFICER, but it can **never** override a Step 2 math breach.
2. Outdated Rulebooks (Stale Thresholds)	Checking today's trade against last year's concentration limit or an expired client carve-out.	Every trade decision stamps the active Mandate ID (IPS-APEX-2026-V4), effective date, and exact character span [char_start..char_end] into the audit record.
3. Data Privacy Leakage (Confidential Holdings)	Exposing full fund positions, prime-broker codes, or internal desk notes to external LLM providers.	**YAML Schema Pruning (fund_schema.yaml):** Strips private columns locally and sends only pre-summed numbers (**240 tokens instead of 3.6M tokens -- a 99.993% cut**).
4. Regulatory Accountability (Who Signs Off Legally)	Under SEC Rule 206(4)-7, legal responsibility rests on a human Chief Compliance Officer (CCO), never an AI.	**Mandatory Officer Sign-Off Queue:** Clear breaches are blocked automatically; qualitative gray areas (TC-09, TC-10) are held until a human officer records written sign-off.

4.1 Testing on 10 Realistic Trades: Accuracy & Money Saved

10 / 10 Benchmark

We tested our prototype (evaluate_10_cases.py and app.py) across **10 realistic proposed orders ($10.70M total order flow)** against a $100.0M settled portfolio using live free-tier models on OpenRouter. A naive single-ticket checker gets only **5/10 (50%)** right and exposes the firm to **-$277,500** in unwind losses and false-alarm costs. Our 4-step gate gets **10/10 (100%)** and saves **+$375,500 ($0 loss paid)**:

ID	Proposed Order & What Makes It Tricky	Basic Check	Our Gate	Loss (No Gate)	Money Saved
TC-01	**Microsoft Corp** ($1.20M AAA Bond) -- Clean trade; rises 3.10% -> 4.30% (<= 5% cap).	APPROVED	APPROVED	$0	$0 (Safe)
TC-02	**Apple Inc** ($1.25M AA+ Bond) -- Direct company breach: rises 4.20% -> 5.45% (> 5% cap).	BLOCKED	BLOCKED	$0	+$45,000
TC-03	**Florida Power & Light** ($1.30M Bond) -- 0% direct, but parent NextEra hits 5.40% (> 5%).	APPROVED (Miss!)	BLOCKED	-$68,000	+$68,000
TC-04	**Ford Motor Credit** ($1.50M Split-Rated BBB-/Ba1) -- Higher rating wins (Sec 2.02b); stays IG.	BLOCKED (False!)	APPROVED	-$82,500	+$82,500
TC-05	**Permian Resources** ($700k BB Energy) -- Total HY is 14.9% (<=15%), but HY Energy hits 5.20% (>5%).	APPROVED (Miss!)	BLOCKED	-$34,000	+$34,000
TC-06	**UK Treasury Gilt** ($1.20M Unhedged GBP Gov Bond) -- Valid OECD government carve-out (8.70% <= 10%).	APPROVED	APPROVED	$0	$0 (Safe)
TC-07	**Siemens AG** ($1.00M Unhedged EUR Corp Bond) -- OECD carve-out is for governments only, not corporates.	BLOCKED	BLOCKED	$0	+$38,000
TC-08	**NVIDIA Corp** ($50k Stock Buy) -- Already at 5.15% from stock gains; holding is fine, buying is banned.	BLOCKED	BLOCKED	$0	+$15,000
TC-09	**Heartland Rail** ($1.00M Bond) -- 4.2% direct coal, but footnote reveals 22% mixed coal rail freight.	APPROVED (Miss!)	HELD (Officer)	-$41,000	+$41,000
TC-10	**Cayman Emerald Trust** ($1.50M SPV Note) -- Footnote reveals orphan SPV backed by oil & gas swap basket.	APPROVED (Miss!)	HELD (Officer)	-$52,000	+$52,000

WHERE THE +$375,500 IN PRE-TRADE PROTECTION COMES FROM (ACROSS 10 SCREENED ORDERS)

4.2 Honest Report: Three Places Where the AI Struggled in Testing

Failure Modes & Fixes

1. **Free-tier reasoning models running out of tokens inside <think> tags:** On our first live OpenRouter test run, a reasoning model spent its entire 500-token budget thinking to itself inside <think>...</think> tags and cut off before outputting JSON. We fixed this by stripping <think> blocks in src/agent_graph.py and routing to fast instruction-tuned models first.
2. **Minor quote paraphrasing on 4 out of 10 live runs:** Smaller free-tier models occasionally dropped a comma or paraphrased a word when copying the rulebook quote. Our Step 4 verifier (verify_verbatim_quote()) caught all 4 instances and automatically anchored the citation to the exact character span [char_start..char_end] from Step 1 so the officer always sees the true legal text.
3. **Free-tier API latency swings (7.5s to 68.6s) & what I would build next:** While Steps 1, 2, and 4 finish locally in **under 5 milliseconds**, waiting on a free-tier LLM for every trade is too slow for a high-speed trading desk. Before scaling this to production, I would enable a **Two-Tier Fast Path**: 90% of standard bond and stock orders have no special deal footnotes or Cayman SPV wrappers, so they can clear Steps 1 & 2 in **< 5 ms**, calling the AI reader only when unstructured rider prose is attached.

A.1 How I Used AI (And Three Prompts That Actually Helped)

One-Page Appendix

I treated AI as a fast sparring partner for legal edge-case brainstorming and UI layout iteration, while keeping all architecture, retrieval math, and compliance state locks in my own hands. Three prompts that were genuinely useful:

1. **Finding real-world compliance traps:** *'Give me concrete examples of how institutional bond funds accidentally violate their SEC Form N-1A mandate even when their basic SQL pre-trade checks on single-issuer % and credit rating pass.'*
2. **Drafting realistic cross-referenced legal clauses:** *'Draft an institutional Investment Policy Statement (IPS) where the 5% single-company cap in Article III depends on a parent-group definition in Article I, a split-rating tie-breaker in Article II, an OECD government bond currency exception in Article IV, and qualitative coal/SPV rules in Article V.'*
3. **Locking down the Step 3 footnote reader prompt:** *'You are a Senior Fund Compliance Auditor. Analyze the proposed trade using ONLY the retrieved IPS clauses, the exact numbers from the Deterministic Calculator, and the deal footnotes. NEVER invent math or override the Deterministic Calculator.'*

A.2 Three AI Suggestions I Rejected (And How I Knew Better)

Human Engineering Judgment

What the AI Suggested	Why I Rejected It & Where My Judgment Came From
1. 'Use a Text-to-SQL agent to query the holdings table dynamically on every trade.'	**How I knew:** At **Apollo Global Management**, I built a natural-language-to-DB assistant over 75K+ holdings and 20K+ trades. For ad-hoc analyst questions, a 98% query accuracy rate is great. For pre-trade compliance, if an LLM forgets GROUP BY ultimate_parent_id even once, the firm executes an illegal trade and pays out-of-pocket unwind losses. I replaced dynamic query generation with fixed, tested aggregation code in Step 2.
2. 'Chunk the rulebook into 500-token slices and use vector embeddings.'	**How I knew:** From building my **Vector Space Retrieval Engine** (benchmarking BM25, TF-IDF, LSA, and WordNet expansion on the Cranfield corpus), I knew dense embeddings struggle with exact section codes (Section 3.01(b)) and sever cross-references when a limit in Article III relies on a definition in Article I. I built a hybrid BM25 + TF-IDF retriever with a 1-hop cross-reference graph expander instead.
3. 'Show the full multi-node backend trace and raw JSON on the main screen.'	**How I knew:** Compliance officers care about two things: *'Can I execute this trade?'* and *'How much loss did we just avoid?'* I stripped all backend jargon from the UI, redesigned it around **AEGIS + Emil Kowalski's** visual principles in a Warm Electric Orange palette, and built dedicated dashboards for **Losses Paid vs. Payments Saved** and **All Activity & Protection**.

A.3 What Came Directly From Me

Practitioner Contributions

- **Framing compliance as Firm P&L Protection (+$375,500 Saved / $0 Loss Paid):** Instead of treating compliance as an abstract pass/fail label, I modeled the real dollar cost of SEC Rule 206(4)-7 trade errors -- quantifying forced-sale unwind slippage (+$167,000), false-alarm spread costs (+$166,500), and remediation fees (+$42,000) across all 10 orders.
- **Schema-pruned context budgeting (data/fund_schema.yaml):** Inspired by my work at Apollo optimizing token budgets across fund schemas, I designed fund_schema.yaml to strip sensitive custodian and broker fields locally and reduce LLM prompt context from ~3.6M tokens to 240 tokens (**99.993% reduction**).
- **Modeling the passive market-drift vs. active purchase lockout (TC-08):** Most prototypes check weight <= 5% statically. I explicitly modeled the transition where holding a drifted position (NVIDIA at 5.15%) is legal, but adding a $50,000 active BUY order is blocked.

A.4 What I Will Happily Defend Live on a Whiteboard Without AI

Live Defense Commitment

With zero AI or notes, I am ready to derive, code, and walk through the entire retrieval and compliance engine live:
1. The Hybrid BM25 + Cross-Reference Graph Retriever (src/retriever.py): Inverted index construction, Okapi BM25 scoring IDF(q) * [tf * (k1 + 1)] / [tf + k1 * (1 - b + b * dl / avgdl)], TF-IDF cosine fusion, 1-hop regex graph expansion across legal sections, and character-offset substring verification.
2. The Pro-Forma Compliance Calculator & Safety Gate (src/deterministic_engine.py): Parent-group relational aggregation on ultimate_parent_id, split-rating ordinal rank resolution (min(rank_SP, rank_Moodys) <= 10), and the one-way safety state machine in Step 4.